DEPARTMENT OF
COMPUTERSCIENCE & ENGINEERING

Experiment No. 10

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Semester: 6th
Subject Name: System Design

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1. Aim

To design a highly scalable online coding judge platform that allows users to write, run, and submit code for algorithmic problems, ensuring efficient execution, accurate evaluation, and reliable handling of high concurrent submissions.

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2. Objective

- Understand the architecture of a large-scale coding judge system.
- Identify functional requirements such as code execution, submission evaluation, and contest handling.
- Identify non-functional requirements including scalability, low latency, and fault tolerance.
- Analyze asynchronous processing using queues for handling large submission loads.
- Design REST APIs for problem management, submissions, and result retrieval.
- Study execution isolation using external services (Judge0) or sandbox environments.

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3. Procedure

- Studied real-world coding platforms and their submission handling workflows.
- Identified core entities such as Users, Problems, Contests, Submissions, and Test Cases.
- Analyzed asynchronous queue-based processing for handling concurrent submissions.
- Collected functional and non-functional requirements for system scalability.
- Designed APIs for problem retrieval, code execution, and submission evaluation.
- Evaluated separation of concerns between API layer, worker services, and execution engine.
- Analyzed usage of external execution services (Judge0) for secure code execution.

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4. System Requirements

A. Functional Requirements

- Users can browse and view coding problems.
- Users can write and execute code with custom input.
- Users can submit code for evaluation against multiple test cases.
- The system evaluates submissions and returns verdicts such as:
  - Accepted
  - Wrong Answer
  - Time Limit Exceeded
  - Runtime Error
  - Compilation Error
- Users can participate in contests and view rankings (optional extension).

B. Non-Functional Requirements

- The system must scale to handle a large number of concurrent submissions.
- The platform must ensure low latency for code execution and result retrieval.
- The system must provide high availability for problem browsing.
- The system must ensure reliability and fault tolerance in submission processing.
- The execution environment must be secure and isolated.

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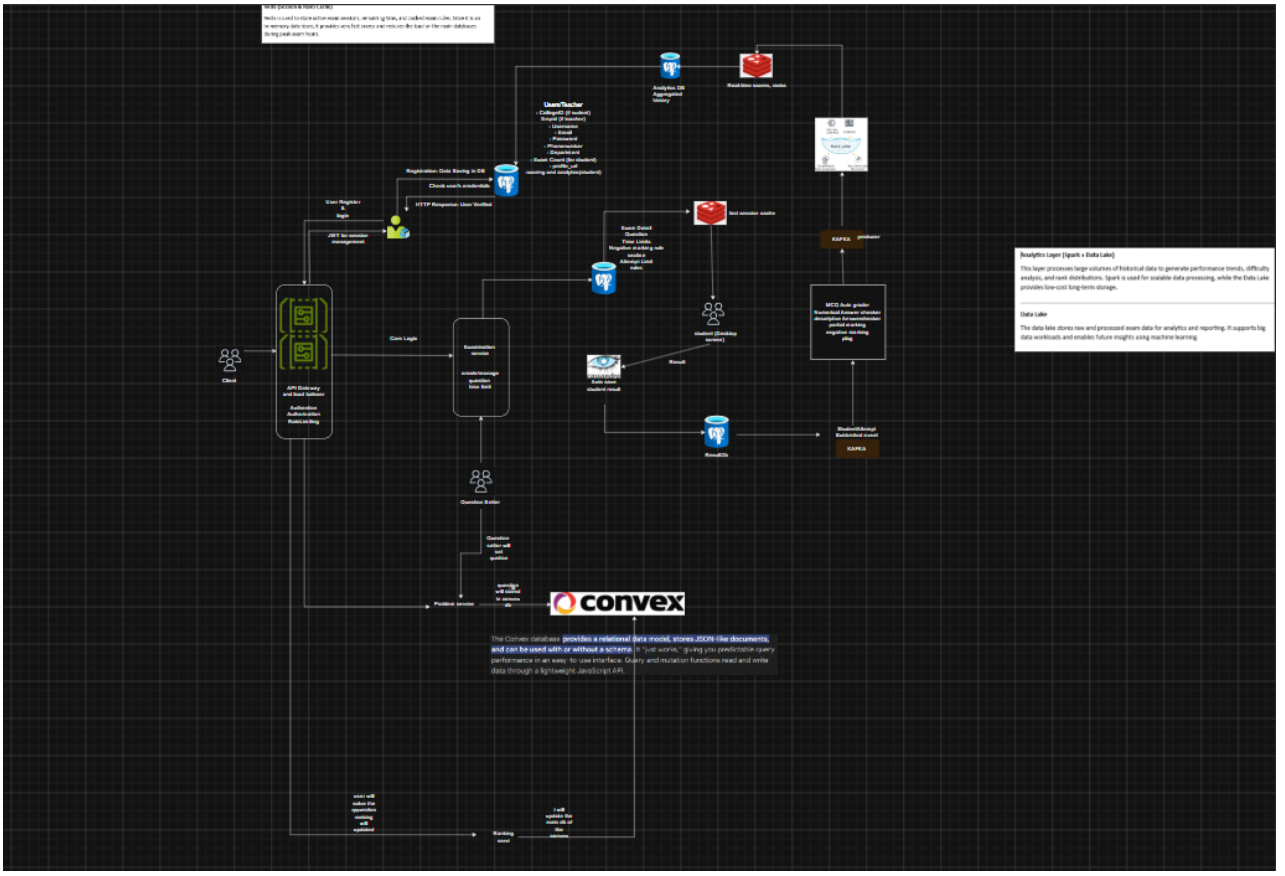
5. High Level Design (HLD)

- **Client Layer**: User interacts via web interface.
- **API Layer**: Handles authentication, problem fetching, and submission requests.
- **Queue System**: Stores submission jobs asynchronously.
- **Worker Layer**: Processes submissions and executes code.
- **Execution Engine**: Uses Judge0 API or sandboxed environment.
- **Database**: Stores users, problems, submissions, and results.
- **Cache Layer**: Stores frequently accessed data like problem metadata.



6. Learning Outcomes (What I Have Learnt)

- Designed a scalable and reliable coding judge system.
- Understood asynchronous processing using queues for handling high traffic.
- Learned how to separate system components for better scalability and maintainability.
- Gained knowledge of secure code execution using external APIs and sandboxing.
- Understood trade-offs between consistency, performance, and scalability.
- Learned how real-world platforms handle distributed execution and high concurrency.